

Morning Paper Report - Tuesday, August 11, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Can working memory be explained by predictive coding?

Mengli Feng, Michael D. Nunez
2025

Source: Crossref; matched terms: working memory, modelling, artificial intelligence, predictive coding

Link: https://doi.org/10.31234/osf.io/5zbc6_v1

Goal: Predictive coding (PC) is a theory in cognitive/computational neuroscience which explains cortical functions with a hierarchical process of minimizing prediction errors

Method: Why is WM hierarchical? Modelling WM in PC frameworks provides alternative explanations to some long-standing questions about WM and may help with resolving the conflicts between WM theories

Results: Since its discovery, PC has been found to be a unifying theory explaining more and more cognitive functions, including perception, attention, and action planning

Conclusion: Why is WM hierarchical? Modelling WM in PC frameworks provides alternative explanations to some long-standing questions about WM and may help with resolving the conflicts between WM theories We expect such alternative explanations can help future researchers, such as in developing computational tools to improve treatments for brain disorders and more robust artificial working memory in artificial intelligence.

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3. Visual working memory load plays limited, to no role in encoding distractor objects during visual search

Mark Lavelle, David Alonso, Roy Luria, Trafton Drew
Visual Cognition, 2021

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: <https://doi.org/10.1080/13506285.2021.1914256>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

4. A Deep Neural Network for Working Memory Load Prediction from EEG Ensemble Empirical Mode Decomposition

Sriniketan Sridhar, Anibal Romney, Vidya Manian

Information, 2023

Source: Crossref; matched terms: working memory, neural network, neural networks, artificial intelligence

Link: <https://doi.org/10.3390/info14090473>

Goal: Mild Cognitive Impairment (MCI) and Alzheimer's Disease (AD) are frequently associated with working memory (WM) dysfunction, which is also observed in various neural psychiatric disorders, including depression, schizophrenia, and ADHD

Method: The data were bandpass-filtered, and independent components analysis was used to select the best electrode channels

Results: Subjects with MCI showed higher values for beta and alpha oscillations in the frontal region than normal subjects

Conclusion: Subjects with MCI showed higher values for beta and alpha oscillations in the frontal region than normal subjects The average power spectral density of the IMFs showed that the IMFs ($p = 0.0469$ for normal subjects and $p = 0.0145$ for subjects with MCI) are robust and reliable features for WM load prediction.

5. Time Regained: Decision Processes in Continuous-Outcome Retrieval from Visual Working Memory

Authors unavailable

2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision

Link: https://osf.io/preprints/psyarxiv/ekvgu_v3/

Goal: Decision-making models of memory retrieval hold that across-trial variability in memory representations and within-trial variability in matching the contents of memory to the retrieval cue both contribute to variability in the speed and accuracy of retrieval decisions

Method: Decision-making models of memory retrieval hold that across-trial variability in memory representations and within-trial variability in matching the contents of memory to the retrieval cue both contribute to variability in the speed and accuracy of retrieval decisions

Results: VWM precision followed either an inverse square-root law or power law in set size, depending on saturation, as predicted by sample-size and attention-weighted sample-size models, respectively

Conclusion: The results support a model in which the capacity limits of VWM reflect the statistical properties of a population of noisy neurons that are divided among items in memory under attentional control and determine the precision of recalled items.